

Technology Stack (Architecture & Stack)

Implement Client Script & UI Policy (Incident)

DATE 01.10.2026	TEAM ID SWTID-2026-5775	PLATFORM ServiceNow	MAX MARKS 4 Marks
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Technical Architecture

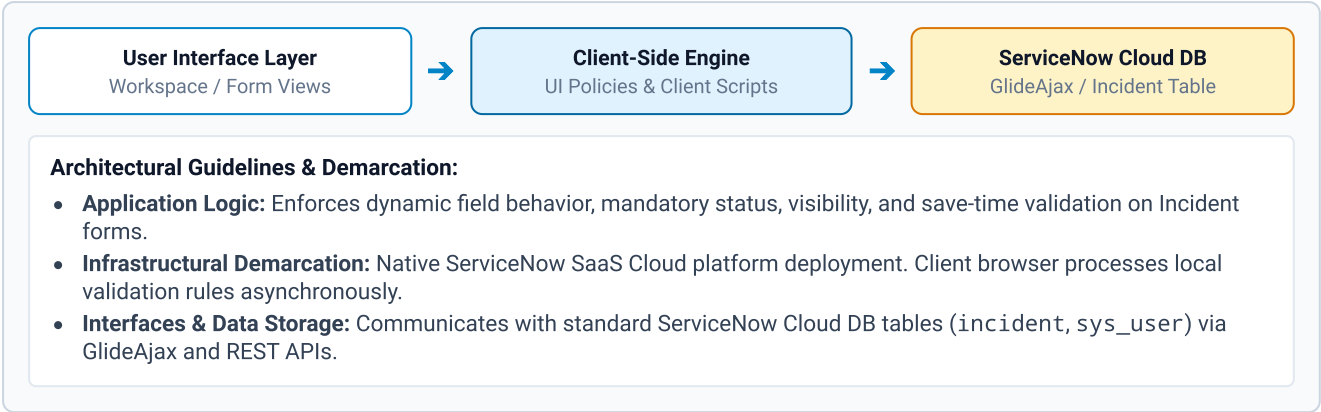


Table 1: Technical Components & Architecture Stack

S.NO	COMPONENT	DESCRIPTION	TECHNOLOGY
1	User Interface	ServiceNow Incident Form Views & Service Operations Workspace.	HTML5, CSS3, JavaScript, GlideForm API
2	Application Logic-1	UI Policy Engine for dynamic field visibility, read-only status, and mandatory rules.	ServiceNow UI Policies & Policy Actions
3	Application Logic-2	Client-side Scripts executing onChange, onLoad, and onSubmit validation logic.	ServiceNow Client Scripts (GlideForm / g_form)
4	Application Logic-3	Asynchronous data fetching for caller profiles and configuration items without form freeze.	GlideAjax & Script Includes (JavaScript)

Table 1: Technical Components (Continued)

S.NO	COMPONENT	DESCRIPTION	TECHNOLOGY
5	Database	Relational database storing Incident records, audit history, and configuration items.	ServiceNow Relational DB Engine (incident table)
6	Cloud Database	High-availability cloud metadata storage for client script definitions and system rules.	ServiceNow Enterprise Cloud Storage
7	File Storage	Attachment storage for screenshots, logs, and evidence attached to Incidents.	ServiceNow Cloud File & Attachment System
8	External API-1	ServiceNow Table API for third-party monitoring tools creating automated incidents.	ServiceNow Table API / REST Services
9	Infrastructure	Enterprise SaaS multi-instance cloud deployment with automated failover.	ServiceNow SaaS Enterprise Cloud

Table 2: Application Characteristics

S.NO	CHARACTERISTICS	DESCRIPTION	TECHNOLOGY
1	Open-Source Frameworks	Standard ECMAScript execution engine supporting client-side scripts.	JavaScript (ES6+ Standard)
2	Security Implementations	Role-based access control (RBAC) and Access Control Lists (ACLs) preventing unauthorized field edits.	ServiceNow ACLs, User Roles (itil, admin), TLS 1.3
3	Scalable Architecture	Client-side execution reduces server load, allowing seamless scaling for high concurrent user volumes.	Multi-Instance Architecture, Client-Side Execution
4	Availability	High availability with redundant cloud infrastructure ensuring continuous platform access.	ServiceNow Advanced High Availability (AHA)
5	Performance	Client scripts optimized to minimize synchronous server calls and optimize form load time.	Asynchronous GlideAjax, DOM-Free Scripting

References:

- ServiceNow Documentation: *Client Scripts & UI Policies Architecture Guide*
- C4 Model Architectural Framework: <https://c4model.com/>
- ServiceNow Developer Docs: *GlideForm (g_form) & GlideAjax API References*